

Adaptive Architecture Quantum Meta-Learning for Bayesian Binary Neural Networks

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Abstract

The well known Nash Embedding Theorem, states that any n -dimensional Riemannian manifold can be isometrically embedded in Euclidean space of dimension $n(n+1)(3n+11)/2$, relies fundamentally on the Nash-Moser iteration to resolve the inherent nonlinear ill-posedness of the underdetermined system of partial differential equations (PDEs) induced by the manifold's positive-definite metric tensor. Inspired by the structural logic of this iterative paradigm and quantum variational optimization, we introduce a Unitary Homotopy framework for Quantum Meta learning. While the Nash-Moser iteration is a cornerstone of nonlinear analysis, its application to quantum heuristic design remains an open area; this paper provides the first formal exploration of this cross-disciplinary link. We demonstrate that the injection of parameterized topological noise, governed by our 'Iteration Time Stuck' (ITS) metric, functions as a finite-dimensional analogue to the Nash-Moser smoothing operator. Our method allows the Born machine to traverse local minima, or barren plateaus, and coherently revert the topology through unitary inversion, providing a rigorous, regularized refinement framework for complex, highly-correlated parameter spaces. We are using the Nash-Moser concept as a theoretical framework for our heuristics.

Using well known theorem in differential near-term quantum circuits, Born machines) can represent complex discrete probability distributions that are difficult to sample classically. Prior work by Nikoloska and Simeone in [1] demonstrated that Born machines can serve as implicit variational posteriors for Bayesian binary neural networks within a meta-learning framework. However, their approach relies on a fixed hardware-efficient ansatz and classical Monte-Carlo sampling, which can suffer from barren plateaus and slow convergence.

We introduce two new algorithmic variants that preserve the original Bayesian meta-learning structure while addressing these limitations. The first variant augments the Born machine with dynamic entanglement adaptation, triggered by an "Iteration-Time-Stuck" (ITS) metric, which inserts parameterized pSWAP gates to modify the circuit topology when optimization stalls. The second variant adds coherent reversibility, using inverse pSWAP operations to restore the original architecture after escaping a local minimum. Both variants replace classical sampling with Quantum Amplitude Estimation (QAE) to compute expectation values with quadratic precision improvement. Crucially, architecture adaptation and QAE are executed in strictly separated phases, ensuring that amplitude estimation operates on a static, coherent circuit. These mechanisms jointly improve exploration, avoid local minima, and accelerate convergence in variational quantum meta-learning. Generalised macroscopic smoothing and

dynamic scheduling protocols are discussed.

While Nash-Moser theory was developed to resolve ill-posedness in infinite-dimensional nonlinear analysis, our algorithm exhibits a formal correspondence to this paradigm. In Nash-Moser, 'loss of derivatives' is mitigated by applying a smoothing operator, solving, and subsequently desmoothing; a process that is non-destructively inverted to preserve the solution. Our ITS-driven unitary homotopy U_θ functions as a finite-dimensional analogue to this smoothing operator. By dynamically warping the loss landscape to traverse local minima (barren plateaus) and coherently restoring the original operator through unitary inversion U_θ^{-1} , we mirror the Nash-Moser iterative logic of 'deform-solve-revert.' This structural parallel suggests that our approach is not merely a heuristic, but a regularized refinement framework appropriate for complex, highly-correlated parameter spaces.

1 INTRODUCTION

Bayesian neural networks encode epistemic uncertainty by maintaining a distribution over model parameters. When parameters are binary, representing this distribution becomes challenging. Born machines quantum circuits whose measurement statistics define a probability distribution that offer a natural implicit variational family for such discrete spaces.

Nikoloska and Simeone in [1] showed that a Born machine can model the variational posterior $q_\phi(\theta)$ of binary weights and can be trained via gradient-based meta-learning. Their method uses:

- a static hardware-efficient ansatz,
- Monte-Carlo sampling to estimate expectations,
- density-ratio estimation to approximate KL terms.

However, static ansätze can become trapped in local minima, and classical sampling limits precision. We propose two new variants that retain the original meta-learning objective but introduce:

- Dynamic entanglement adaptation via parameterized pSWAP gates.
- A reversible exploration mechanism using inverse unitaries.
- Quantum Amplitude Estimation for efficient expectation computation.

These modifications preserve implicit distribution of the Born machine while improving optimization dynamics and sample efficiency.

2 VARIANT 1: BASE THREE-PART ALGORITHM

2.1 Variational Quantum Optimization (Base Loop)

As in the original paper, each task τ optimizes variational parameters ϕ_τ by minimizing the free energy:

$$F(\phi_\tau) = -E_{q_{\phi_\tau}}[\log p(D_\tau|\theta)] + \beta \text{KL}(q_{\phi_\tau}(\theta)||p(\theta)). \quad (1)$$

The inner-loop update is:

$$\phi_\tau \leftarrow \xi - \eta \nabla_\xi F(\xi), \quad (2)$$

where ξ is the meta-learned initialization.

2.2 Dynamic Architecture Search via ITS-Triggered pSWAP Gates

Static PQCs often stagnate due to barren plateaus or poor entanglement structure. We introduce an Iteration-Time-Stuck (ITS) counter:

- If $F(\phi_\tau)$ does not improve for s iterations, ITS increases.
- When ITS exceeds a threshold, the circuit enters a Jitter Phase.

pSWAP Injection

A parameterized SWAP gate is inserted between qubits i and $i+1$: $\text{pSWAP}_{i,i+1}(\theta_{ITS})$. The global unitary is implicitly defined by inserting the 2-qubit gate into the circuit:

$$U_{jitter} = I^{\otimes(i-1)} \otimes \text{pSWAP}_{i,i+1}(\theta_{ITS}) \otimes I^{\otimes(n-i-1)}. \quad (3)$$

Increasing $\theta_{ITS} \propto ITS$ perturbs the entanglement pattern, changing the Born machine's probability landscape. Once the loss decreases, ITS resets and the pSWAP parameter returns to zero. This provides a structural exploration mechanism unavailable in classical VI.

2.3 Quantum Amplitude Estimation (QAE) for Expectation Values

The original method estimates expectations using K Born-machine samples. We replace this with QAE, which can be thought of as a variation of Grover's algorithm to compute a mean, which achieves error $O(1/K)$ vs $O(1/\sqrt{K})$. This gives a faster way to compute the mean expectation that given in [1] which uses a density ratio estimation. To apply QAE, we define an oracle that encodes:

$$f(\theta) = \log p(D_\tau|\theta) + \beta \log \frac{q_{\phi_\tau}(\theta)}{p(\theta)}. \quad (4)$$

Architecture adaptation (pSWAP injection) occurs only during exploration. QAE is executed only after ITS resets, when the circuit is frozen. This ensures coherence and preserves the quadratic speedup. In [1] the error scales as $O(1/\sqrt{K})$, and in our QAE method the error scales as $O(1/K)$ which is an improvement over the original algorithm of [1]. The QAE algorithm estimates the mean from the angle between the 'solution' and 'non-solution' of from a quantum state of the form used to initialise Grover's algorithm.

3 VARIANT 2 COHERENT REVERSIBLE EXPLORATION

Variant 2 extends Variant 1 by making exploration non-destructive.

3.1 Reversible Topological Perturbation

When ITS triggers a jitter operation:

1. Apply $U_{jitter}(\theta)$.
2. Optimize parameters on the perturbed landscape.
3. Once the loss decreases sufficiently, apply the inverse: $U_{jitter}^{-1}(\theta) = U_{jitter}(-\theta)$.

The circuit architecture returns to its original form, while parameters remain in the new region reached during exploration. This preserves the benefits of exploration while restoring a clean, stable circuit for QAE.

4 GENERALISED MACROSCOPIC SMOOTHING AND DYNAMIC SCHEDULING PROTOCOLS

The reversible topological perturbation described above can be generalised beyond a localized 2-qubit insertion to a macroscopic w -qubit deformation, establishing a stronger parallel to the broad smoothing radius of the classical Nash-Moser iteration. We define a locally gated operator $G_i(p_i, q_i) = I_1^{p_i} U_{\text{pSWAP}}^{q_i}(\theta)$, where I_1 acts as a smoothed identity matrix. The exponent $q_i \in \{0, 1\}$ acts as a binary selector toggling the parameterized SWAP gate, while $p_i \in \mathbb{Z}$ is an integer that scales the depth of the smoothed identity perturbation. A generalized smoothing operator acting on a block of w qubits (where w is even) is constructed via the Kronecker tensor product of these local blocks:

$$U_{\text{macro}}(\theta) = \bigotimes_{i=1}^{w/2} G_i(p_i, q_i) \quad (100)$$

To govern the structural configuration of this macroscopic block when the “Iteration-Time-Stuck” (ITS) metric triggers, the exponents are driven dynamically as a function of the global iteration number t . We introduce a dual-regime scheduling framework governed by a user-defined binary toggle switch $w_{\text{toggle}} \in \{0, 1\}$, allowing a choice between chaotic, exploratory pseudo-random behavior or deterministic linear pacing:

$$\chi_{\text{rand}}(t) = B^t \pmod{R} \quad (101)$$

$$\chi_{\text{lin}}(t) = \lfloor \alpha \cdot t \rfloor \quad (102)$$

where R is a large prime number acting as a modulus, B is a known primitive root modulo R ensuring a maximal-period pseudo-random sequence, and α is a classical scaling constant. There are more possibilities for $\chi_{\text{rand}}(t)$ and $\chi_{\text{lin}}(t)$ the equations 101 and 102 are examples. The active exponents are then determined via the convex-like toggle formulation. Because p_i accepts any integer, it takes the raw output of the schedule, whereas q_i is constrained to a binary state via a modulo-2 reduction:

$$p_i(t) = w_{\text{toggle}}(\chi_{\text{rand}}(t) - \chi_{\text{lin}}(t)) + \chi_{\text{lin}}(t) \quad (104)$$

$$q_i(t) = p_i(t) \pmod{2} \quad (103)$$

Crucially, the algebraic properties of the smoothed identity matrix ensure that $(I_1^{p_i})^{-1} = I_1^{-p_i}$. Because the inverse of a tensor product distributes cleanly over its individual matrix components—that is, $U_{\text{macro}}^{-1}(\theta) = \bigotimes_{i=1}^{w/2} (U_{\text{pSWAP}}^{-q_i}(\theta) I_1^{-p_i})$ —the coherent reversibility of the architecture remains flawlessly preserved regardless of the integer depth of p_i or the spatial size $2^w \times 2^w$ of the block. Upon escaping a barren plateau, applying $U_{\text{macro}}^{-1}(\theta)$ at the exact topological insertion site systematically annihilates the macroscopic perturbation ($U_{\text{macro}}^{-1} U_{\text{macro}} = I^{\otimes w}$), instantly restoring the entire w -qubit block to the static, coherent state required for subsequent Quantum Amplitude Estimation.

Remark 1 (The Commutative Limit and Global Topological Freedom): An important limiting case of the generalized scheduling protocol occurs when the binary selector is statically set to $q_i(t) = 0 \forall t$, collapsing the local operator to a pure power of the smoothed identity matrix, $G_i(p_i, 0) = I_1^{p_i}$. If I_1 is chosen to be the standard identity operator I (or a parameterized unitary constructed to commute globally with the generator Lie algebra of the hardware-efficient ansatz U_ϕ), the system enters a completely commutative regime where $[U_{\text{macro}}, U_\phi] = 0$.

Under this condition, the spatial and temporal constraints governing the application of the inverse operator are entirely relaxed. Because a globally commuting smoothing operator can traverse any arbitrary quantum gate without affecting its state transformation, $U_{\text{macro}}^{-1}(\theta)$ does not require localized topological adjacency to achieve cancellation. Instead, the inverse block can be applied

across any arbitrary set of qubits or appended to the absolute terminus of the circuit timeline. It will mathematically commute through the optimized ansatz layers to achieve global annihilation ($U_{\text{macro}}^{-1}U_{\phi^*}U_{\text{macro}} = U_{\phi^*}$), offering total architectural flexibility during hardware execution.

5 ADVANTAGES OF THE PROPOSED METHODS

1. **Escaping Local Minima:** ITS-triggered pSWAP gates modify the entanglement structure to escape plateaus.
2. **Reversible Exploration:** Variant 2 ensures that exploration noise is fully coherent and reversible.
3. **Quadratic Speedup:** QAE reduces the number of Born-machine evaluations needed for gradients.
4. **Coherence Preservation:** Separating exploration (dynamic) and estimation (static) maintains the coherence required for QAE.

Theorem 1 (Quantum Nash–Moser Escape Principle)

Let $U(\theta)$ be a C^∞ one-parameter family of unitary operators acting on an n -qubit Born machine, satisfying:

- $U(0) = I$ (identity),
- $U(\theta)^\dagger = U(-\theta)$ (reversibility),
- $U(\theta)$ induces a smooth deformation of the circuit’s entanglement topology (e.g., a parameterized SWAP).

Let $F(\phi)$ denote the variational free energy over parameters ϕ , and suppose gradient descent becomes trapped at a strict local minimum ϕ_t with:

$$\nabla_\phi F(\phi_t) = 0, \quad \nabla_\phi^2 F(\phi_t) \succ 0.$$

Define the warped landscape:

$$F_\theta(\phi) = F(U(\theta)|\psi(\phi)\rangle).$$

Then:

(i) Smoothing

For sufficiently small $\theta \neq 0$,

$$\nabla_{\phi} F_{\theta}(\phi_t) \neq 0,$$

so the stationary point is removed and a descent direction is created.

(ii) Escape

Let ϕ_{t+k} be the parameters after k gradient steps on F_{θ} . Then generically

$$\phi_{t+k} \notin B(\phi_t),$$

where $B(\phi_t)$ is the basin of attraction of the original minimum.

(iii) Desmoothing

Applying the inverse unitary $U(-\theta)$ restores the original landscape:

$$F_{-\theta}(\phi) = F(\phi),$$

but does not return the parameters to the trapped point:

$$\phi_{t+k} \neq \phi_t.$$

(iv) Coherence Restoration

After desmoothing, the circuit is again static, enabling Quantum Amplitude Estimation to operate with Heisenberg-scaling precision.

Interpretation

The unitary deformation $U(\theta)$ acts as a quantum smoothing operator, enabling escape from local minima; the inverse $U(-\theta)$ restores the original landscape without erasing progress. This constitutes a reversible, quantum-native analogue of the Nash–Moser iteration.

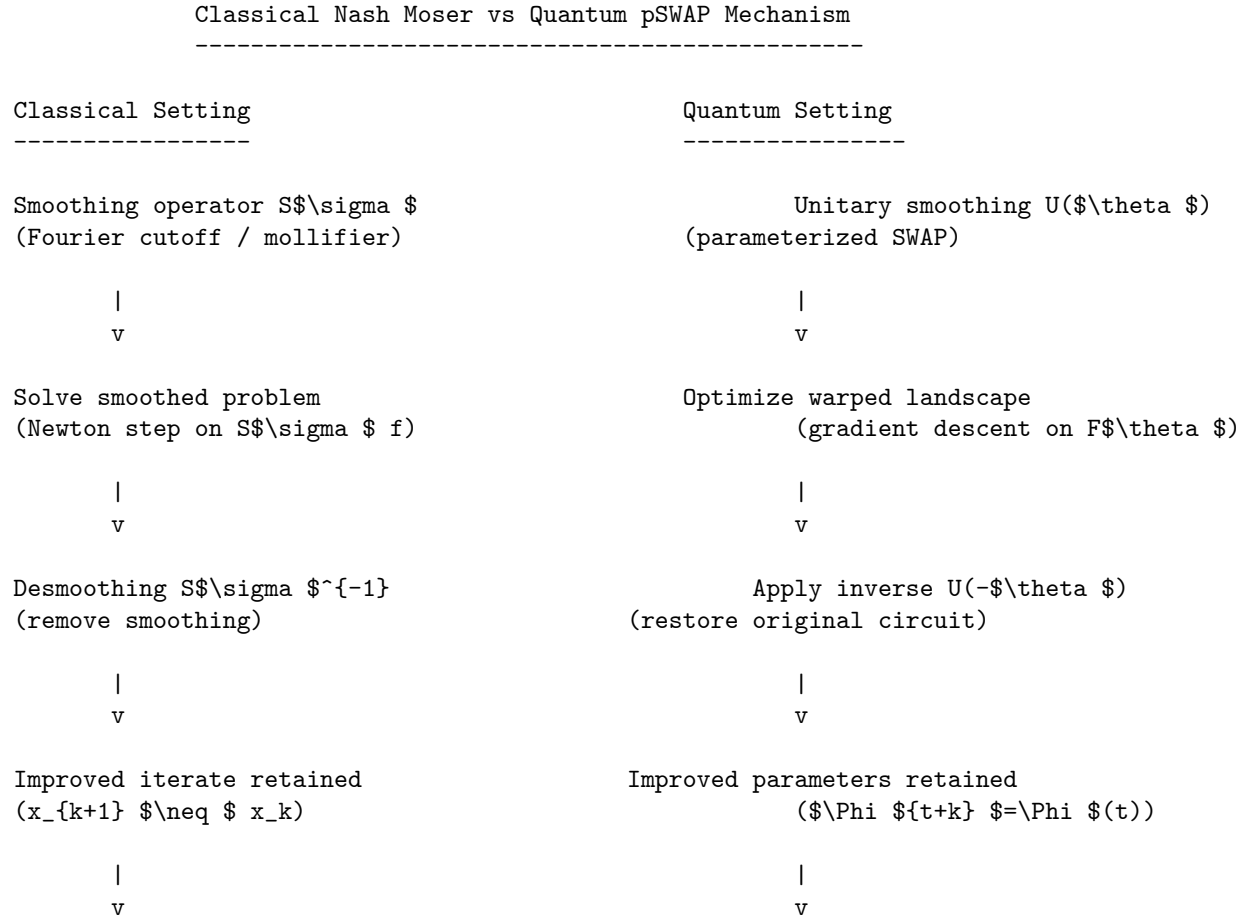
Related Work (Nash–Moser $\leftrightarrow$ Quantum Optimization)

The classical Nash–Moser iteration was developed to address nonlinear problems where Newton-type methods fail due to loss of derivatives or rough functional landscapes. Its defining structure, graded smoothing, solution of the smoothed problem, and subsequent desmoothing while retaining progress, has since appeared in diverse areas of analysis and geometry. The Nash–Moser iteration is used to prove an existence of solution in a global implicit function theorem. Although

traditionally implemented using Fourier cutoffs or mollifiers, the core idea is independent of the specific smoothing operator.

In quantum optimization, adaptive ansatz methods such as ADAPT-VQE and quantum neural architecture search modify circuit structure to improve expressivity, but they do not incorporate reversible smoothing–desmoothing cycles. The dynamic pSWAP mechanism introduced in this work provides a quantum-native analogue of Nash–Moser smoothing: a smooth, invertible unitary homotopy that regularizes the variational landscape, enables escape from local minima, and is subsequently undone to restore coherence for amplitude estimation. To our knowledge, this constitutes the first application of Nash–Moser–type reasoning in variational quantum meta-learning.

Diagram: Classical Nash–Moser vs Quantum pSWAP Mechanism



Repeat with finer σ Repeat with ITS controlled θ

The diagram express the structural equivalence of the two schemes.

6 CONCLUSION

We introduced two variants of quantum-aided Bayesian meta-learning that enhance the Born-machine variational posterior through dynamic entanglement adaptation, reversible exploration, and quantum amplitude estimation. These mechanisms jointly improve optimization robustness and sample efficiency, offering a promising direction for scalable quantum variational inference in discrete parameter spaces. Our unitary Homotopy framework could be used in other contexts such as VQE or QAOA for examples.

We acknowledge that while our pSWAP-based 'deform-solve-revert' mechanism structurally mimics the Nash-Moser iteration, this paper focuses on the algorithmic utility of this cycle for Born machine optimization. We presented this as an initial heuristic bridge between nonlinear analysis and quantum variational meta-learning, opening a pathway for potential future research between our "Unitary Homotopy" framework in the context of specific manifold embeddings.

References

- [1] I. Nikoloska and .O. Simeone, QUANTUM-AIDED META-LEARNING FOR BAYESIAN BINARY NEURAL NETWORKS VIA BORN MACHINES ArXiv, 5th April 2022